SECOND-LARGEST AND MINIMUM DISTANCE MULTIPLICITIES IN PLANAR POINT SETS: AN UPPER BOUND BELOW 3/2, AND RELATED RESULTS ON ERDŐS PROBLEM #132

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ABSTRACT. For a finite set $X \subset \mathbb{R}^2$ let Δ_2 and δ be the second-largest and the smallest distance determined by X, and let $\mu(d)$ be the number of pairs of X at distance d. Clemen, Dumitrescu and Liu asked for $L = \limsup_{n\to\infty} \max_{|X|=n} \min\{\mu(\Delta_2), \mu(\delta)\}/n$; the known bounds were $9/7 \le L \le 3/2$, the upper bound coming from Vesztergombi's inequality $\mu(\Delta_2) \le 3n/2$. We prove that $\min\{\mu(\Delta_2), \mu(\delta)\} \le \frac{54}{37}n + C_0$ for an absolute constant C_0, so $L \le 54/37 < 3/2$.

We also give two results on Erdős Problem #132, which asks whether some distance other than the diameter must occur at most n times. First, if no vertex of the convex hull has four points of X at distance Δ_2 (for instance, if no four points of X are concyclic), then $\mu(\Delta_2) \le \frac{3}{2}|L_1| + |L_2|$, where L_1, L_2 are the first two convex layers; so, under this hypothesis, the answer is yes whenever at least half as many points lie at depth ≥ 3 as on the hull. This rests on a layer bound whose inputs from Vesztergombi's structure theory we prove directly. Second, for n points in convex position there are at least three distances occurring between 1 and n times when n is even and at least 6, and when $n \in \{7, 9, 11, 13\}$. The cases $n = 11, 13$ are computer-assisted: they reduce to the unsatisfiability of finite ordinal problems, certified by DRAT/LRAT proof checkers, and for $n = 7, 11, 13$ also by a Lean 4 formalisation.

AI DISCLOSURE

The author used an AI assistant (Claude, Anthropic) for exploration, computation, formalisation and drafting. The checks performed on the results are described in Appendix A.

1. INTRODUCTION

Let $X \subset \mathbb{R}^2$ be a set of n points. Write $\Delta = \Delta(X)$ for its diameter, Δ_2 for its second-largest distance and δ for its minimum distance, and write $\mu(d) = \mu_X(d)$ for the number of unordered pairs of points of X at distance d. Hopf and Pannwitz [14] proved that $\mu(\Delta) \le n$.

Erdős Problem #132. Erdős and Pach asked whether there must be a *second* distance, besides the diameter, that occurs at least once and at most n times, and whether the number of such distances must tend to infinity (perhaps being at least $n^{1-o(1)}$). This is Problem #132 on the Erdős problems website [4], where it is recorded as open and where Erdős's offer of \$100 for "any nontrivial result" is recorded (reference [Er97e] there). The statement fails for $n = 4$ (two equilateral triangles glued along an edge) and holds for $n = 5, 6$ by Erdős and Fishburn [9]. We call a distance d *rare* if $1 \le \mu(d) \le n$.

Clemen, Dumitrescu and Liu [5] proved that the answer to the first question is yes for sets in convex position (their Theorem 1.2), and that $\mu(\Delta_2) \le n$ whenever

$$\min\Big\{\tfrac{3}{2}(|L_1| + |L_2|),\ \tfrac{4}{3}|L_1| + 2|L_2|,\ 2|L_1| + |L_2|\Big\} \le n, \tag{1}$$

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where L_1, L_2 are the first two convex layers (their Theorem 1.3). They showed that neither Δ_2 nor δ is always rare: they constructed sets with $\min\{\mu(\Delta_2), \mu(\delta)\} \geq 9n/8 - o(n)$ (their Proposition 1.5). They asked (their Problem 1.6) for the value of

$$L := \limsup_{n\to\infty} \max_{|X|=n} \frac{\min\{\mu(\Delta_2), \mu(\delta)\}}{n}.$$

Vesztergombi [19] proved that $\mu(\Delta_2) \leq \frac{3}{2}n$ for every n-point planar set, which gives $L \leq 3/2$. On the Erdős problems forum, the user ienjoymath [15] gave a construction showing $L \geq 9/7$ and conjectured $L = 9/7$. As far as we know, $9/7 \leq L \leq 3/2$ were the best bounds before this paper.

Results. Our main result is an upper bound strictly below 3/2.

Theorem 1. *There is an absolute constant C_0 such that every n-point set $X \subset \mathbb{R}^2$ satisfies*

$$\min\{\mu(\Delta_2), \mu(\delta)\} \;\leq\; \tfrac{54}{37}\, n + C_0.$$

Consequently $L \leq 54/37 \approx 1.4595$.

(Here X is assumed to determine at least two distances; otherwise $n \leq 3$.)

The constant C_0 is explicit; the proof gives $C_0 = 3N_0 + 10^8$ with $N_0 = 1.51 \cdot 10^9$ (Section 7). We have made no attempt to optimise it.

Theorem 8.6 and Corollary 8.7 give a new class of sets for which the answer to the first question of Problem #132 is yes. Let $m_3 := n - |L_1| - |L_2|$ be the number of points of depth at least 3. Condition (1) contains the case $2|L_1| + |L_2| \leq n$, i.e. $|L_1| \leq m_3$. We show that $|L_1| \leq 2m_3$ suffices under a genericity hypothesis that holds, for instance, when no four points of X are concyclic. The inputs we need from Vesztergombi's structure theory [20] are proved here from scratch (Proposition 8.2 and Theorem 8.5).

Section 9 treats the second question of Problem #132 in convex position. Theorem 9.1 states that a set of n points in convex position has at least three rare distances for every n with $6 \leq n \leq 13$. Some of these cases follow from the literature, some follow from a parity-sharp version of the counting argument of [5], and the cases $n = 11, 13$ are new and computer-assisted. In Section 10 we list open problems.

Idea of the proof of Theorem 1. Let S be the set of points that have a partner at distance Δ_2. Vesztergombi's bound applied to S gives $\mu(\Delta_2) \leq \frac{3}{2}|S|$. Every δ-pair not inside S has an endpoint outside S, and a point has at most 6 neighbours at distance δ, so $\mu(\delta) \leq e(S) + 6(n - |S|)$, where $e(S)$ is the number of δ-pairs inside S. A linear combination of these two bounds (Lemma 3.3) shows that any bound $e(S) \leq k|S| + O(1)$ with $k < 3/2$ beats 3/2; with $k = 4/3$ it gives 54/37. Almost all points of S lie on the boundary of a convex body: either the "ball polygon" $K = \bigcap_{w\in X} \mathrm{B}(w, \Delta_2)$, or the convex hull. Each such point has a well-defined inward direction. We show that all but $O(1)$ of the δ-edges inside S join points with almost equal inward directions. Such edges come in two kinds: tangential edges, and *rungs* joining the two boundaries, whose geometry is fixed by an exact identity (Lemma 2.2(c)). An ownership/pairing argument then bounds the density of rungs. The regime where the diameter is much larger than Δ_2 is handled separately by an incidence argument.

Notation. $\mathrm{B}(c, r)$ and $\mathrm{C}(c, r)$ are the closed disc and the circle with centre c and radius r. For unit vectors u, u', $\angle(u, u') \in [0, \pi]$ is the angle between them; recall $|u - u'| \leq \angle(u, u')$. Angles are in radians unless marked with a degree sign.

2. Set-up for the proof of Theorem 1

We scale so that $\delta = 1$. Throughout Sections 2 to 7, X is an n-point set with $n \geq 4$ and at least two distinct distances, so $\Delta > \Delta_2 \geq \delta = 1$. We put

$$t := \Delta - \Delta_2 > 0, \qquad \tau := t - \frac{1-t^2}{2\Delta_2}, \qquad \beta := \frac{1}{100}.$$

Then $\tau - t = (t^2-1)/(2\Delta_2)$. Hence $\tau \leq t$ when $t \leq 1$, and $\tau > 1$ if and only if $t > 1$.[C2] A crucial feature of X is that it has *no distance in the open interval* (Δ_2, Δ).

We use the following objects.

- D is the set of endpoints of diametral pairs. Each point of D is a vertex of $P := \operatorname{conv} X$.
- S is the set of points of X that have a partner at distance Δ_2; $R := X \setminus S$ and $r := |R|$, $s := |S|$.
- $K := \bigcap_{w \in X} \mathrm{B}(w, \Delta_2)$, a compact convex set (a "ball polygon").
- $e(Y)$ is the number of δ-pairs inside $Y \subseteq X$.

The *degenerate regime* is $\Delta > 1.94\,\Delta_2$ (Section 6); the *nondegenerate regime* is $\Delta \leq 1.94\,\Delta_2$ (Sections 3 to 5).

Lemma 2.1 (ball polygon)**.** *(a)* $X \setminus D \subseteq K$.
(b) $S \setminus D \subseteq \partial K$.
(c) $\operatorname{dist}(z, K) \geq t$ *for every* $z \in D$.

Proof. (a) A point $y \notin D$ has all its distances at most Δ_2, since there are none in (Δ_2, Δ). (b) If $v \in S \setminus D$ and $|vw| = \Delta_2$, then $v \in K \cap \mathrm{C}(w, \Delta_2) \subseteq \partial K$. (c) If $z \in D$ and $|zx| = \Delta$, then $K \subseteq \mathrm{B}(x, \Delta_2)$, while $|zx| = \Delta_2 + t$. □

Direction field. For each $p \in S$ we fix a unit vector u_p.

- If $p \in S \setminus D$, choose a Δ_2-partner w_p and put $u_p := (w_p - p)/\Delta_2$. Since $K \subseteq \mathrm{B}(w_p, \Delta_2)$, u_p is an inward normal of K at p.
- If $p \in S \cap D$, choose a diametral partner x_p and put $u_p := (x_p - p)/\Delta$. This is an inward normal of P at p.

Lemma 2.2 (exact identities)**.** *Let* $y \in X$, $y \neq p$.

(a) If $p \in S \cap D$, *then* $(y-p)\cdot u_p \geq |y-p|^2/(2\Delta) > 0$.
(b) If $p \in S \setminus D$, *then either* $(y-p)\cdot u_p \geq |y-p|^2/(2\Delta_2)$ *(we say* y *is* normal *for* p*), or* $|y - w_p| = \Delta$, *so that* $y \in D$ *(we say* y *is an* exception *for* p*).*
(c) In the exception case with $|y-p| = 1$ *we have exactly* $(y-p)\cdot u_p = -\tau$. *Hence* $y - p = -\tau u_p \pm \sigma u_p^{\perp}$ *with* $\sigma := \sqrt{1-\tau^2}$. *In particular exceptions at distance* 1 *exist only when* $\tau \leq 1$.

Proof. (a) $|y - x_p|^2 = |y-p|^2 - 2\Delta(y-p)\cdot u_p + \Delta^2 \leq \Delta^2$. (b) $|y - w_p|^2 = |y-p|^2 - 2\Delta_2(y-p)\cdot u_p + \Delta_2^2$, and $|y - w_p| \in [0, \Delta_2] \cup \{\Delta\}$. (c) With $|y-p| = 1$ and $|y - w_p| = \Delta = \Delta_2 + t$ the same identity gives $(y-p)\cdot u_p = (1 + \Delta_2^2 - \Delta^2)/(2\Delta_2) = (1 - 2\Delta_2 t - t^2)/(2\Delta_2) = -\tau$.[C2] □

Lemma 2.3 (no common normal)**.** *If* $p \neq q$ *both lie in* $S \setminus D$, *then* $u_p \neq u_q$.

Proof. Neither point lies in D, so each is normal for the other: $(q-p)\cdot u_p > 0$ and $(p-q)\cdot u_q > 0$. These cannot both hold if $u_p = u_q$. □

Good edges, T-edges and rungs. A δ-edge pq inside S is *good* if $\angle(u_p, u_q) \le \beta$, and *bad* otherwise. For a good edge put $e := q - p$, $a := e \cdot u_p$ and $b := -e \cdot u_q$. Then $|a + b| = |e \cdot (u_p - u_q)| \le \beta$.

Lemma 2.4 (dichotomy)**.** *Assume $\tau > 2\beta$. Each good edge is of exactly one of the following two kinds.*

- T-edge: $a, b \in (0, \beta)$.
- Rung: *one end $v \in S \setminus D$ sees the other end $z \in S \cap D$ as an exception. Then $(z-v)\cdot u_v = -\tau$ and $(v - z) \cdot u_z \in [\tau - \beta, \tau + \beta]$.*

Proof. By Lemma 2.2, from each end the other end is either normal (then the corresponding one of a, b is positive) or an exception (then it equals $-\tau$). An exception target lies in D, and points of D see everything as normal, so both ends cannot be exceptions. If neither is, then $a, b > 0$ and $a + b \le \beta$. If one is, say $a = -\tau$, then $b \in [\tau - \beta, \tau + \beta]$, and $b > 0$ shows that b is not an exception value. □

We say a T-neighbour q of p is on the *right* (resp. *left*) of p if $(q - p) \cdot u_p^{\perp} > 0$ (resp. < 0), where $u^{\perp}$ denotes u rotated by $-90°$.

(T1) Each p has at most one T-neighbour on each side. Indeed, two T-directions on one side both make an angle in $(0, \arcsin \beta)$ with $\pm u_p^{\perp}$, so they are less than $60°$ apart, whereas two points at distance 1 from p and at distance ≥ 1 from each other are at least $60°$ apart as seen from p.

(T2) If q is the right T-neighbour of p then p is the left T-neighbour of q, since $\angle(u_p, u_q) \le \beta$.

Hence the T-edges form vertex-disjoint directed paths and cycles, and every cycle has length at least 3 (a vertex cannot be both the left and the right T-neighbour of another).

Turning. Let γ be a closed arc of the boundary of a compact convex set with nonempty interior. Its *turning* $\operatorname{turn}(\gamma)$ is the length of the set $N(\gamma) \subseteq S^1$, the union of the normal cones at *all* points of γ, including the full normal cones at its endpoints.

Lemma 2.5 (multiplicity)**.** *Let $\gamma_1, \dots, \gamma_m$ be closed boundary arcs of one convex body, and suppose every boundary point lies in at most M of them. Then $\sum_i \operatorname{turn}(\gamma_i) \le 2\pi M$.*

Proof. For all but countably many directions $\nu \in S^1$ the set of boundary points with outer (or inner) normal ν is a single point y_ν, and $\nu \in N(\gamma_i)$ if and only if $y_\nu \in \gamma_i$. Integrate over ν. □

Points of $S \setminus D$ lie on ∂K and points of $S \cap D$ lie on ∂P; in each case distinct points are at Euclidean, hence arc-length, distance at least 1. So a boundary arc of length ℓ contains at most $\ell + 1$ of them, and a boundary of perimeter ℓ contains at most ℓ of them.

3. Reductions

Lemma 3.1 ((VS))**.** *For every $S' \subseteq S$, the number of Δ_2-pairs inside S' is at most $\frac{3}{2}|S'|$. In particular $\mu(\Delta_2) \le \frac{3}{2}s$.*

Proof. Every Δ_2-pair of X lies in S, and every distance within S' is a distance of X. If S' contains no Δ_2-pair there is nothing to prove. If S' contains a diametral pair of X, then $\operatorname{diam} S' = \Delta$ and, because X has no distances in (Δ_2, Δ), Δ_2 is the second-largest distance of S'; Vesztergombi's theorem [19] gives at most $\frac{3}{2}|S'|$ such pairs. Otherwise every distance in S' is at most Δ_2, so $\Delta_2 = \operatorname{diam} S'$ and Hopf–Pannwitz [14] gives at most $|S'|$. □

We use Vesztergombi's theorem in exactly the form of its main theorem: for any set of m points in the plane, the second-largest distance occurs at most $\frac{3}{2}m$ times [19, Theorem]. There is no threshold on m and no additive constant.

Lemma 3.2 ((E))**.** $\mu(\delta) \le e(S) + 6r$.

Proof. Every δ-pair not inside S has an endpoint in R, and each point has at most 6 points at distance δ. □

Lemma 3.3. *Suppose* $e(S) \le ks + C$ *with* $0 \le k < 3/2$. *Then* $\min\{\mu(\Delta_2), \mu(\delta)\} \le \frac{9}{7.5-k}\, n + C$.

Proof. Put $\lambda = (6-k)/(7.5-k) \in (0,1)$. By Lemmas 3.1 and 3.2,

$$\min\{\mu(\Delta_2), \mu(\delta)\} \le \lambda \cdot \tfrac{3}{2}s + (1-\lambda)(ks + C + 6r) \le \tfrac{9}{7.5-k}(s + r) + C.$$

For $k = 1, \frac{5}{4}, \frac{4}{3}$ the factor $9/(7.5-k)$ is $\frac{18}{13}, \frac{36}{25}, \frac{54}{37}$.[C1] □

Lemma 3.4. *Let* $S_1 \subseteq S$ *consist of points of* Δ_2*-degree exactly* 1, *with* $p_1 := |S_1|$. *Suppose* $e(S) \le s + 2p_1 + C$. *Then* $\min\{\mu(\Delta_2), \mu(\delta)\} \le \frac{7}{5}n + C$.

Proof. At most p_1 Δ_2-pairs touch S_1, and the rest lie inside $S \setminus S_1$, so $\mu(\Delta_2) \le p_1 + \frac{3}{2}(s - p_1)$ by Lemma 3.1. Hence $\min \le \frac{4}{5}(\frac{3}{2}s - \frac{1}{2}p_1) + \frac{1}{5}(s + 2p_1 + C + 6r) = \frac{7}{5}s + \frac{6}{5}r + \frac{C}{5} \le \frac{7}{5}n + C$.[C1] □

So in the nondegenerate regime it suffices to show $e(S) \le \frac{4}{3}s + O(1)$ in every τ-regime except one, and to show that the remaining regime satisfies the hypothesis of Lemma 3.4.

Small cases. Suppose we are in the nondegenerate regime and $\Delta_2 < 10^4$. Then $\Delta < 1.94 \cdot 10^4$, so X lies in a disc of radius Δ about any of its points, and since the discs of radius $\frac{1}{2}$ about points of X are disjoint, $n \le (2\Delta + 1)^2 < N_0 := 1.51 \cdot 10^9$.[C11] Since $\mu(\delta) \le 3n$, Theorem 1 holds for such X once $C_0 \ge 3N_0$. *From now on, in the nondegenerate regime,* $\Delta_2 \ge 10^4$; in particular $2/\Delta_2 \le \beta/2$.[C3]

4. Bad edges are $O(1)$ in the nondegenerate regime

4.1. **A lens lemma.**

Lemma 4.1 (lens)**.** *Let* $\Lambda = \mathrm{B}(c_1, \rho) \cap \mathrm{B}(c_2, \rho)$ *with* $0 < |c_1c_2| = d < 2\rho$, *and put* $\cos\varphi_0 := d/(2\rho)$. *Assume*

$$\rho \;\ge\; \frac{2}{\cos(\varphi_0/2) - \cos\varphi_0}. \tag{2}$$

Let p *lie on the arc of* $\partial\Lambda$ *belonging to* $\mathrm{C}(c_1, \rho)$, *let* q *lie on the arc belonging to* $\mathrm{C}(c_2, \rho)$, *and suppose* $|pq| \le 1$. *Then there is a corner* c *of* Λ *such that the boundary arcs from* p *to* c *and from* c *to* q *each have length at most* $\rho_0 := 1/\sin(\varphi_0/2)$.

Proof. Take $c_1 = (-d/2, 0)$, $c_2 = (d/2, 0)$, $p = c_1 + \rho(\cos\varphi, \sin\varphi)$ and $q = c_2 + \rho(-\cos\psi, \sin\psi)$ with $|\varphi|, |\psi| \le \varphi_0$. *Step 1.* $p_x - q_x = \rho(\cos\varphi - \cos\varphi_0) + \rho(\cos\psi - \cos\varphi_0)$ with both terms ≥ 0; since $|p_x - q_x| \le 1$, each term is at most 1. By (2), $\cos\varphi < \cos(\varphi_0/2)$, i.e. $|\varphi| > \varphi_0/2$; likewise $|\psi| > \varphi_0/2$. *Step 2.* By the mean value theorem, $\rho(\varphi_0 - |\varphi|) = \rho(\cos\varphi - \cos\varphi_0)/\sin\xi \le 1/\sin\xi$ for some $\xi \in (\varphi_0/2, \varphi_0) \subseteq (0, \pi/2)$, so $\rho(\varphi_0 - |\varphi|) \le \rho_0$; likewise for ψ. *Step 3.* If φ, ψ had opposite signs, then $|p_y - q_y| \ge 2\rho\sin(\varphi_0/2)$. Now (2) gives $\rho \ge 2/(1 - \cos\varphi_0) = 1/\sin^2(\varphi_0/2)$, so $2\rho\sin(\varphi_0/2) \ge 2/\sin(\varphi_0/2) \ge 2 > 1$, a contradiction. *Step 4.* So the corner $c = (0, \mathrm{sign}(\varphi)\rho\sin\varphi_0)$ works. □

Lemma 4.2 (lens-corner arc length)**.** *In the situation of Lemma 4.1, suppose moreover that* $p, q \in \partial K'$ *for a compact convex* $K' \subseteq \Lambda$, *and* $|pq| = 1$. *Let* H *be the closed half-plane bounded by the line* pq *that contains* c, *and let* $\gamma := \partial K' \cap H$ *be the boundary arc of* K' *from* p *to* q *on* c*'s side. Then* $\mathrm{length}(\gamma) \le 2\rho_0$.

Proof. $\partial\Lambda$ contains no segment, so the line pq meets it only in p and q, and $\partial\Lambda \cap H$ is exactly the arc $p \to c \to q$, of length at most $2\rho_0$. Since $K' \cap H \subseteq \Lambda \cap H$ are convex, monotonicity of perimeter gives $\text{length}(\gamma) + 1 \le 2\rho_0 + 1$. (If $K' \cap H$ is the segment $[p, q]$, then γ is that segment, of length $1 \le 2\rho_0$.) □

Constants.[C7] For *KK lenses* ($\rho = \Delta_2$, $d \le \Delta \le 1.94\Delta_2$) we have $\cos\varphi_0 \le 0.97$, so $\rho_0^2 = 2/(1-\cos\varphi_0) \le 66.7 < 8.2^2$. The function $c \mapsto \cos(\varphi_0/2) - c = \sqrt{(1+c)/2} - c$ is decreasing on $[0, 1)$, so (2) holds as soon as $\rho \ge 89.01$, in particular for $\rho = \Delta_2 \ge 10^4$. For *DD lenses* ($\rho = \Delta$, $d \le \Delta$) we have $\cos\varphi_0 \le \frac{1}{2}$, $\rho_0 \le 2$, and (2) needs only $\rho \ge 5.5$.

4.2. **The count.** We split bad edges pq by type: *KK* (both ends in $S \setminus D$), *DD* (both in $S \cap D$) and *cross* (one end in each).

If the perimeter of K is below 1202, then $|S \setminus D| < 1202$ and at most $6 \cdot 1202$ edges touch $S \setminus D$; similarly for P and $S \cap D$. We may therefore assume both perimeters are at least 1202 when counting the bad edges touching the corresponding row; these small cases contribute a constant below $1.5 \cdot 10^4$ in total.

KK edges. Let $p, q \in S \setminus D$ form a bad edge. If $w_p = w_q$, then $\angle(u_p, u_q) \le 2\arcsin(1/(2\Delta_2)) < \beta$,[C3] so the edge would be good; hence $w_p \ne w_q$. Then $K \subseteq \Lambda := \mathrm{B}(w_p, \Delta_2) \cap \mathrm{B}(w_q, \Delta_2)$, p lies on the $\mathrm{C}(w_p, \Delta_2)$-arc and q on the $\mathrm{C}(w_q, \Delta_2)$-arc of $\partial\Lambda$, and $d = |w_p w_q| \le \Delta \le 1.94\Delta_2$. Lemmas 4.1 and 4.2 give an arc $\gamma_{pq} \subseteq \partial K$ from p to q of length at most $2\rho_0$. The union of the normal cones along the closed arc γ_{pq} is connected and contains u_p and u_q, so $\text{turn}(\gamma_{pq}) \ge \angle(u_p, u_q) > \beta$. If $y \in \gamma_{pq}$ then p lies within arc distance $2\rho_0$ of y; there are at most $4\rho_0 + 1$ such p, each of degree at most 6, so every point lies on at most $M = 6(4\rho_0 + 1)$ of the arcs. By Lemma 2.5,

$$\#\text{KK-bad} \le 12\pi(4\rho_0 + 1)/\beta < 1.28 \cdot 10^5.\text{[C9]}$$

DD edges. The same argument, on ∂P with the lens $\mathrm{B}(x_p, \Delta) \cap \mathrm{B}(x_q, \Delta)$ and $\rho_0 \le 2$, gives $\#\text{DD-bad} \le 12\pi \cdot 9/\beta < 3.4 \cdot 10^4$. (If $x_p = x_q$ the angle is less than $2/\Delta < \beta$ and the edge is good.)

Cross edges. Let zv be a bad edge with $z \in S \cap D$ and $v \in S \setminus D$. Since $z \in X$, $|zw_v| \in [0, \Delta_2] \cup \{\Delta\}$.

(c1) $|zw_v| = \Delta$. By Lemma 2.2(a), $(w_v - z)/\Delta$ is a second inward normal of P at z. Its angle to u_v is the angle at w_v of the triangle zvw_v; that angle is opposite the side of length 1, hence acute, and by the law of sines it is at most $\arcsin(1/\Delta) < 2/\Delta_2 \le \beta/2$.[C3] As $\angle(u_z, u_v) > \beta$, the normal cone of P at z has angle greater than $\beta/2$. Normal cones at distinct vertices have disjoint interiors, so fewer than $4\pi/\beta$ points z qualify, each of degree at most 6: $\#(\text{c1}) < 24\pi/\beta < 7600$.

(c2) $|zw_v| \le \Delta_2$. Let g be the unit vector from v towards x_z, and $\alpha := \angle(u_v, g)$.

(i) *Depth.* $v \notin D$ gives $|vx_z| \le \Delta_2$, while $|zx_z| = \Delta$ and $|zv| = 1$ give $|vx_z| \ge \Delta - 1$. So $v \in \mathrm{B}(x_z, \Delta_2)$ at depth $d_0 := \Delta_2 - |vx_z| \in [0, 1-t]$; in particular this case needs $t \le 1$. Also $K \subseteq \mathrm{B}(x_z, \Delta_2)$.

(ii) *α is not small.* $\angle(u_z, g)$ is the angle at x_z of the triangle zvx_z, which is at most $\arcsin(1/\Delta) < \beta/2$; hence $\alpha \ge \angle(u_v, u_z) - \angle(u_z, g) > \beta/2$.

(iii) *α is not close to π.* Since $w_v, x_z \in X$, $|w_v x_z| \le \Delta \le 1.94\Delta_2$. The law of cosines in the triangle vw_vx_z (sides Δ_2 and $\Delta_2 - d_0$ at v, enclosing the angle α) gives $\cos\alpha \ge (1 + \varrho^2 - 1.94^2)/(2\varrho)$, where $\varrho := (\Delta_2 - d_0)/\Delta_2 \in [1 - 10^{-4}, 1]$. The right side is increasing in ϱ, so $\cos\alpha \ge -0.8821$, i.e. $\alpha \le 151.9°$. By concavity of sin on $[0, \pi]$, $\sin\alpha \ge \min\{\sin(\beta/2), 0.471\} = \sin(\beta/2)$.[C8]

(iv) *The walk.* Let $e_0 \perp u_v$ be the unit vector with $e_0 \cdot g = -\sin\alpha < 0$. Follow ∂K from v in the direction whose forward one-sided tangent is closest to e_0, parametrised by arc length, $\gamma(0) = v$. Suppose the turning of $\gamma([0,\sigma])$ is at most $\beta/4$. This turning includes the whole normal cone $N_K(v)$. Then every normal along the arc is within $\beta/4$ of u_v, so every forward tangent γ' is within $\beta/4$ of e_0 (at a corner v, the forward tangent is perpendicular to an extreme normal of $N_K(v)$, which is also within $\beta/4$ of u_v). Hence $\gamma'\cdot g \le -\sin\alpha + 2\sin(\beta/8) \le -\sin(\beta/2) + 2\sin(\beta/8) < -\beta/5$.[C8] Integrating, $(\gamma(\sigma)-v)\cdot g \le -\sigma\beta/5$, and so

$$|\gamma(\sigma) - x_z|^2 \;\ge\; (\Delta_2 - d_0)^2 + \tfrac{2}{5}(\Delta_2 - d_0)\,\sigma\beta.$$

For $\sigma = L := 6/\beta = 600$ the right side exceeds Δ_2^2,[C8] which puts $\gamma(L)$ outside $\mathrm{B}(x_z, \Delta_2) \supseteq K$, a contradiction. Hence the closed arc J_v of length L from v in the chosen direction has turning greater than $\beta/4$.

(v) *Count.* A point y lies in J_v only if v is within arc distance L of y, which allows at most $2L+1$ points v. By Lemma 2.5, $\#\{v\} \le 2\pi(2L+1)/(\beta/4)$. Each such v has degree at most 6, so $\#(\mathrm{c2}) \le 48\pi(2L+1)/\beta < 1.82 \cdot 10^7$.[C9]

Proposition 4.3. *In the nondegenerate regime the number C_{bad} of bad edges satisfies $C_{\mathrm{bad}} < 1.83 \cdot 10^7$. Together with the constants 71 and 12 of Section 5, $C_{\mathrm{bad}} + 71 + 12 < 10^8$.*[C9]

5. The nondegenerate regime

5.1. **Region I: $\tau \le \sqrt{3}/2 - 2\beta$.** Let $p \in S \setminus D$ and let q be a good neighbour of p; the *elevation* of $q - p$ is $\arcsin((q-p)\cdot u_p)$. If q is normal for p, then $b > 0$ (by Lemma 2.2(a) if $q \in D$, by (b) if $q \notin D$), so $a < \beta$. Otherwise q is an exception and $a = -\tau$.

- If $\tau \ge 0$, every good neighbour on one side of $\mathbb{R}u_p$ has elevation in $[-\arcsin\tau, \arcsin\beta)$, a window of width less than $\arcsin(\sqrt{3}/2 - 2\beta) + \arcsin\beta \approx 58.35^\circ < 60^\circ$.[C4]
- If $\tau < 0$, then $|\tau| \le 1/(2\Delta_2)$ and all elevations lie in $(0, \arcsin\beta) \cup \{\arcsin|\tau|\}$; the window is again less than 60°.[C4]

Let $p \in S \cap D$. Then $a > 0$ always, and either $a < \beta$, or p is the exception of q and $a \in [\tau-\beta, \tau+\beta]$; the window is at most $\arcsin(\sqrt{3}/2 - \beta) \approx 58.87^\circ < 60^\circ$.[C4]

Since two neighbours of p are at least 60° apart, p has at most one good neighbour on each side. Hence $e(S) \le s + C_{\mathrm{bad}}$, and Lemma 3.3 with $k = 1$ gives $\min\{\mu(\Delta_2), \mu(\delta)\} \le \frac{18}{13}n + C$.

5.2. **The regions with $\tau > \sqrt{3}/2 - 2\beta$.** Here $\tau > 2\beta$,[C5,C6] so Lemma 2.4 applies. *Region II* is $\tau \ge \sqrt{3}/2 + 3\beta$; *Region III* is $\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$.

Region II with $\tau > 1$. Then $t > 1$, and by Lemma 2.2(c) there are no rungs. By (T1) the T-edges number at most s, so $e(S) \le s + C_{\mathrm{bad}}$, which gives $\frac{18}{13}n + C$.

From now on in Region II assume $\sqrt{3}/2 + 3\beta \le \tau \le 1$. Then $t \le 1$ and $t \ge \tau$, so $t - \arcsin\beta \ge 0.886$; also $\sigma = \sqrt{1-\tau^2} \le \sigma_{\max} \approx 0.4440$.[C5]

5.2.1. *Mixed T-edges.*

Lemma 5.1. *Assume $\sqrt{3}/2 + 3\beta \le \tau \le 1$. Let $v \in S \setminus D$ and $z \in S \cap D$ be joined by a T-edge. Then the closed arc of ∂K of length 2 starting at v on z's side has turning at least $t - \arcsin\beta$.*

Proof. Suppose the turning is $\Theta < t - \arcsin\beta$; note $t - \arcsin\beta < 1 < \pi/3$. Let e_0 be the unit tangent direction at v perpendicular to u_v pointing to z's side. Every forward tangent along the arc is within Θ of e_0, so $|\gamma(\sigma) - v| \ge \sigma\cos\Theta \ge \sigma/2$, and some $\sigma^* \le 2$ has $|\gamma(\sigma^*) - v| = 1$. The unit vector $\gamma(\sigma^*) - v$ lies within Θ of e_0, and the unit vector $z - v$ lies within $\arcsin\beta$ of e_0. Hence $|\gamma(\sigma^*) - z| \le \Theta + \arcsin\beta < t \le \mathrm{dist}(z, K)$ by Lemma 2.1(c). But $\gamma(\sigma^*) \in K$, a contradiction. □

Each v has at most two T-edges, and v must lie within arc distance 2 of a point y of its arc, so at most 5 such v and $M \le 10$. By Lemma 2.5 the number of mixed T-edges is at most $2\pi \cdot 10/0.886 < 71$.[C5]

From here on, the *K-row* is $S \setminus D$ and the *D-row* is $S \cap D$, each with its own *pure* T-edges; rungs join the two rows.

5.2.2. *Rungs.* For a rung (v, z) write $z = v + o_{\sigma'}(u_v)$, where $o_\pm(u) := -\tau u \pm \sigma u^\perp$ (Lemma 2.2(c)). The map $u \mapsto o_\pm(u)$ is a rotation, so $|o_\pm(u) - o_\pm(u')| = |u - u'| \le \angle(u, u')$; and $|o_+(u) - o_-(u)| = 2\sigma$. Hence $|o_\pm(u) - o_\pm(u')| \le 2\sigma + \angle(u, u')$ for any choice of signs.

(R1) *Each vertex has at most one rung.* At v: the two candidate positions $v + o_\pm(u_v)$ are $2\sigma < 1$ apart. At z: if z had rungs to $v \ne v'$, both rungs are good, so $\angle(u_v, u_{v'}) \le 2\beta$ and $|v - v'| \le 2\sigma + 2\beta \approx 0.908 < 1$,[C5] a contradiction.

T-steps. If q is the right T-neighbour of p, then $q - p = \cos\varepsilon\, u_p^\perp + \sin\varepsilon\, u_p$ with $\varepsilon \in (0, \arcsin\beta)$. So for T-steps e_1 at v_0 and e_2 at z_0, $\angle(e_1, e_2) \le \angle(u_{v_0}, u_{z_0}) + \arcsin\beta$.

Lemma 5.2 ((R2), proximity)**.** *Let (v_0, z_0) be a rung, v_1 the right pure T-neighbour of v_0 and z_1 the right pure T-neighbour of z_0; put $e_1 = v_1 - v_0$, $e_2 = z_1 - z_0$. If v_1 has a rung (v_1, z'), then $z' = z_1$.*

Proof. $z' - z_1 = (e_1 - e_2) + (o_{\sigma_1}(u_{v_1}) - o_{\sigma_0}(u_{v_0}))$. Since $\angle(u_{v_0}, u_{z_0}) \le \beta$ (good rung) and $\angle(u_{v_0}, u_{v_1}) \le \beta$ (good T-edge), $|z' - z_1| \le (\beta + \arcsin\beta) + \beta + 2\sigma \approx 0.918 < 1$.[C5] Distinct points of X are at least 1 apart. □

Lemma 5.3 ((R2-sym))**.** *In the setting of Lemma 5.2, if z_1 has a rung (v', z_1), then $v' = v_1$.*

Proof. $v' = z_1 - o_{\sigma'}(u_{v'})$ and $v_1 = z_0 - o_{\sigma_0}(u_{v_0}) + e_1$, so $v' - v_1 = (e_2 - e_1) - (o_{\sigma'}(u_{v'}) - o_{\sigma_0}(u_{v_0}))$. The angle between $u_{v'}$ and u_{v_0} is at most 3β, as the sum of three good-edge angles along $v' \to z_1 \to z_0 \to v_0$. Hence $|v' - v_1| \le (\beta + \arcsin\beta) + 3\beta + 2\sigma \le 2\sigma + 6\beta \approx 0.948 < 1$.[C5] □

(R3) *Parallel rungs and rhombi.* Rungs with the same vector c satisfy $c \cdot u_v = -\tau$, so u_v is one of the two unit solutions, which lie at $\pm\theta_0$ from $-c$, where $\theta_0 := \arccos\tau \in [0, \pi/2]$. Rungs form a matching by (R1), so their K-ends are distinct and by Lemma 2.3 have distinct normals. Hence *no three rungs are parallel*, and two parallel rungs have normals exactly $2\theta_0$ apart. If v_0v_1, v_1z_1, z_1z_0, z_0v_0 are all unit edges (a *rhombus*), then v_1 and z_0 are the two points of $\mathrm{C}(v_0, 1) \cap \mathrm{C}(z_1, 1)$, so $v_0 + z_1 = v_1 + z_0$ and the rungs (v_0, z_0) and (v_1, z_1) are parallel. Since the T-edge v_0v_1 is good, $2\theta_0 = \angle(u_{v_0}, u_{v_1}) \le \beta$. So *rhombi exist only if* $\tau \ge \cos(\beta/2)$, and $\cos(\beta/2) > \sqrt{3}/2 + 3\beta$.[C5]

5.2.3. *Ownership and pairing.* For a rung $\mathsf{r} = (v, z)$, walk right from v along pure K-row T-edges until reaching the first vertex after v that is a rung vertex, or until the walk stops. Let $k \ge 1$ be the number of vertices visited before stopping, v included. Define d in the same way in the D-row, starting from z. Then r *owns* these $k + d$ vertices. Every other row vertex is *unowned*: it has no rung vertex to its left on its T-path or T-cycle. Ownership is well defined, since each owned vertex belongs to the nearest rung vertex at or to its left, and that rung vertex lies on exactly one rung.

Charging. Charge to r the rung itself and the right T-edge out of each vertex it owns. Charge every other pure T-edge to its unowned left endpoint. Every rung and pure T-edge is charged exactly once, and every vertex has at most one right T-edge. Hence:

- $\mathrm{edges}(\mathsf{r}) \le 1 + k + d$, with equality only if both walks end at a rung vertex ("both reach"); otherwise $\mathrm{edges}(\mathsf{r}) \le k + d$;
- unowned vertices carry at most one edge each.

Both-reach gaps are $(1,1)$ *or* $(\geq 2, \geq 2)$. If both walks reach and $k = 1$, then z_0 has a right neighbour z_1 and v_1 is a rung vertex; by Lemma 5.2 its rung is (v_1, z_1), so $d = 1$. If both reach and $d = 1$, then v_0 has a right neighbour v_1 and z_1 is a rung vertex; by Lemma 5.3 its rung is (v_1, z_1), so $k = 1$. A $(1,1)$ gap is a rhombus $v_0v_1z_1z_0$ (by (R1), $z_1 \neq z_0$ and $v_1 \neq v_0$). A cycle with a single rung vertex gives a both-reach gap with k equal to the cycle length, which is at least 3.

Ratios. A rhombus gap has ratio edges/vertices $= 3/2$; a both-reach gap with $k, d \geq 2$ has ratio at most $5/4$; every other gap has ratio at most 1.

If $\tau < \cos(\beta/2)$ there are no rhombi, so $e(S) \leq \frac{5}{4}s + C_{\text{bad}} + 71$ and Lemma 3.3 gives $\frac{36}{25}n + C$.

If $\tau \geq \cos(\beta/2)$, pair each rhombus rung $\mathsf{r}_0 = (v_0, z_0)$ with its successor $\mathsf{r}_1 = (v_1, z_1)$. The rung r_1 is not itself a rhombus rung: otherwise $\mathsf{r}_0, \mathsf{r}_1, \mathsf{r}_2$ would be three parallel rungs (here $v_2 \neq v_0$ because T-cycles have length at least 3). The pairing is injective, because v_0 is the unique left T-neighbour of v_1. If r_1 owns m vertices, the pair has ratio at most

$$\max\Big\{\frac{4+m}{2+m} : m \geq 4\Big\} \cup \Big\{\frac{3+m}{2+m} : m \geq 2\Big\} = \frac{4}{3}.^{[C5]}$$

Hence $e(S) \leq \frac{4}{3}s + C_{\text{bad}} + 71$, and Lemma 3.3 gives $\frac{54}{37}n + C$. This is the only place where $54/37$ is attained.

5.3. **Region III:** $\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$**.** Here $\theta_0 = \arccos\tau \in (26.36°, 32.22°)$.

Lemma 5.4 (corner lemma)**.** *Suppose $v \in S \setminus D$ has a rung partner z and a second Δ_2-partner $w' \neq w_v$, and put $u' := (w' - v)/\Delta_2$. Then $\angle(u_v, u') \geq \min\{2\theta_0,\ 90° - \theta_0\} > 2\pi/7$.*

Proof. Put $c = z - v$. If $|zw'| = \Delta$, then Lemma 2.2(c) applied with w' gives $c \cdot u' = -\tau$, so u' is the other unit solution and the angle is $2\theta_0$. If $|zw'| \leq \Delta_2$, then $c \cdot u' > 0$ while $\angle(c, u_v) = 180° - \theta_0$, so the angle is at least $90° - \theta_0$. Numerically, $2\theta_0 > 2\pi/7$ because $\tau < \sqrt{3}/2 + 3\beta < \cos(\pi/7)$, and $90° - \theta_0 > 2\pi/7$ because $\tau > \sqrt{3}/2 - 2\beta > \cos(3\pi/14)$.[C6] □

Both u_v and u' are inward normals of K at v, and $\angle(u_v, u') < \pi$ because $|w_v w'| \leq \Delta < 2\Delta_2$. So the normal cone $N_K(v)$ has angle greater than $2\pi/7$, and at most 6 such v exist. Every other K-vertex that has a rung has Δ_2-degree exactly 1; let S_1 be the set of these vertices. A K-vertex has at most 2 rungs (the two positions $v + o_\pm(u_v)$), and (T1) bounds all T-edges, mixed ones included, by s. Hence

$$e(S) \leq s + 2|S_1| + 12 + C_{\text{bad}},$$

and Lemma 3.4 gives $\frac{7}{5}n + C$.

Remark 5.5. This regime genuinely needs the weighted Lemma 3.4: a "hexagonal strip" around a single centre has average δ-degree 3 locally.

6. The degenerate regime $\Delta > 1.94\Delta_2$

Proposition 6.1. *If $\Delta > 1.94\,\Delta_2$ and $n \geq 4$, then $\mu(\Delta_2) \leq n + 1$.*

Put $\kappa := (\Delta_2/\Delta)^2 < 1/1.94^2$. Note that this section does not use δ.

Step 1: no two disjoint diametral pairs. Suppose (x, x') and (z, z') are disjoint diametral pairs with midpoints m, m'. The four cross distances satisfy $\sum |ab|^2 = 2\Delta^2 + 4|m - m'|^2 \geq 2\Delta^2$, and each is either at most Δ_2 or equal to Δ. If at most one equals Δ, the sum is at most $\Delta^2(1 + 3\kappa) < 2\Delta^2$,[C10] a contradiction. So at least two cross distances equal Δ.

- If two disjoint cross pairs equal Δ, say $|xz| = |x'z'| = \Delta$, then z and x' are the two points of $\mathrm{C}(x, \Delta) \cap \mathrm{C}(z', \Delta)$, so x, z, z', x' form a rhombus of side Δ whose diagonals satisfy $d_1^2 + d_2^2 = 4\Delta^2$. One diagonal is at least $\sqrt{2}\Delta$, a contradiction.

- If two cross pairs sharing a point equal Δ, say $|xz| = |xz'| = \Delta$, then xzz' is an equilateral triangle of side Δ, and $x' \in \mathrm{C}(x, \Delta)$ with $|x'z|, |x'z'| \le \Delta$ lies on the 60° arc of $\mathrm{C}(x, \Delta)$ between z and z', strictly inside it. One of $|x'z|, |x'z'|$ is at least $2\Delta \sin 15^\circ$, and $(2 \sin 15^\circ)^2 = 2 - \sqrt{3} > \kappa$,[C10] so that distance lies in (Δ_2, Δ), which is impossible.

Step 2: the diameter graph is a star. A graph with no two disjoint edges is a star or a triangle. A triangle is impossible for $n \ge 4$: a fourth point is not diametral, so it is within Δ_2 of all three vertices, but $\Delta_2^2 < \Delta^2/3$ is less than the squared circumradius.[C10] So $D = \{y\} \cup A$ with A the set of diametral partners of y; put $B := X \setminus D \subseteq K$ and $W := \{y\} \cup A$.

Step 3: where the Δ_2-pairs are.

- *Within B: none.* For $a \in A$, $K \subseteq \mathrm{B}(y, \Delta_2) \cap \mathrm{B}(a, \Delta_2)$, a lens whose bounding box has squared diagonal less than $0.24\,\Delta_2^2$.[C10] So $\operatorname{diam} K < \Delta_2$.
- *Within A: at most one.* $A \subseteq \mathrm{C}(y, \Delta)$, and its pairwise distances are at most Δ_2 (a Δ-pair inside A would create a triangle in the diameter graph). A chord of $\mathrm{C}(y, \Delta)$ of length at most $\Delta_2 < \Delta/1.94$ subtends an angle less than 30° at y. So A lies in an arc of angular width less than 60°, on which chord length is increasing in angular separation; a Δ_2-pair in A must realise the largest pairwise distance, hence be the two extreme points of A.
- *Between y and A: none*, as those distances equal Δ.

Hence $\mu(\Delta_2) \le 1 + I$, where $I := \#\{(b, w) : b \in B,\ w \in W,\ |bw| = \Delta_2\}$.

Step 4: $I \le |B| + |W|$. We may assume K has nonempty interior: an open chord between two points of K lies in the interior of every disc $\mathrm{B}(w, \Delta_2)$, so otherwise $|B| \le |K \cap X| \le 1$ and the bound is trivial. For $w \in X$ let $J_w := \mathrm{C}(w, \Delta_2) \cap K$.

- *Each J_w is a single closed arc (possibly a point).* For $x \in X \setminus \{w\}$, $\mathrm{C}(w, \Delta_2) \cap \mathrm{B}(x, \Delta_2)$ is the closed arc $\{\cos(\theta - \theta_x) \ge |wx|/(2\Delta_2)\}$, of angular length at most π. For $p, q \in J_w$ we have $|pq| < \Delta_2 < 2\Delta_2$, so p, q are not antipodal and the minor arc of $\mathrm{C}(w, \Delta_2)$ from p to q lies in every $\mathrm{B}(x, \Delta_2)$, hence in K. Since $\operatorname{diam} K < \Delta_2$, the angular spread of J_w is less than 60°, so J_w is a single closed arc.
- *No point is interior to two arcs.* Suppose b is a relative-interior point of a nondegenerate arc J_w and $b \in \mathrm{C}(w', \Delta_2)$ with $w' \in X \setminus \{w\}$. The function $f(\theta) = |w + \Delta_2 e(\theta) - w'|^2 - \Delta_2^2 = |w - w'|^2 + 2\Delta_2\, e(\theta) \cdot (w - w')$ is ≤ 0 on J_w (as $K \subseteq \mathrm{B}(w', \Delta_2)$) and vanishes at b, so it has an interior local maximum with value 0. But every local maximum of this shifted cosine is its global maximum $|w - w'|^2 + 2\Delta_2 |w - w'| > 0$, a contradiction.
- *Consequences.* Each nondegenerate J_w is a subarc of ∂K, and distinct nondegenerate arcs have disjoint relative interiors. So a point $b \in B$ lies on at most two nondegenerate W-arcs (one on each side); any further incidences of b come from degenerate arcs $J_w = \{b\}$.
- *Counting.* Let I_{nd} be the number of incidences from nondegenerate arcs. Then $I_{\mathrm{nd}} \le |B| + \#\{b : \text{two } W\text{-arcs meet at } b\}$. Mapping each such b to the W-arc that starts at b in the clockwise direction is injective, so $I_{\mathrm{nd}} \le |B| + \#\{w : J_w \text{ nondegenerate}\}$. Each degenerate arc contributes at most one incidence. Hence $I \le |B| + |W|$.

Therefore $\mu(\Delta_2) \le 1 + |B| + |A| + 1 = n + 1$, proving Proposition 6.1.

7. Assembly of the proof of Theorem 1

The τ-intervals partition $\mathbb{R}$, the split at $\Delta = 1.94\,\Delta_2$ is exhaustive, and the cases $n \le 3$ (or X with a single distance) are trivial. Every constant in the last column is at most $54/37$,[C1] and $n + 1 \le \frac{54}{37} n + 1$. With $C < 10^8$ from Proposition 4.3, Theorem 1 holds with $C_0 = 3N_0 + 10^8$. □

Remark 7.1. The two least standard parts of the argument are the ownership/pairing bookkeeping in Section 5 and the incidence count in Step 4 of Section 6. Both were re-derived independently, and both were also checked by computer in abstract models (Appendix A). The

regime ($\delta = 1$, $\beta = 1/100$)	bound on $e(S)$	bound on $\min\{\mu(\Delta_2), \mu(\delta)\}$
degenerate: $\Delta > 1.94\Delta_2$, $n \geq 4$	—	$n+1$ (Proposition 6.1)
nondegenerate, $\Delta_2 < 10^4$	—	$3n < 3N_0$ (Section 3)
I: $\tau \leq \sqrt{3}/2 - 2\beta$ (incl. $\tau \leq 0$)	$s + C$	$\frac{18}{13}n + C$
III: $\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$	$s + 2\lvert S_1\rvert + C$	$\frac{7}{5}n + C$
II: $\sqrt{3}/2 + 3\beta \leq \tau < \cos(\beta/2)$	$\frac{5}{4}s + C$	$\frac{36}{25}n + C$
II: $\cos(\beta/2) \leq \tau \leq 1$	$\frac{4}{3}s + C$	$\frac{54}{37}n + C$
II: $\tau > 1$ ($\Leftrightarrow t > 1$)	$s + C$	$\frac{18}{13}n + C$

TABLE 1. The regimes in the proof of Theorem 1; here $C < 10^8$.

bookkeeping check is exhaustive up to $6+6$ row vertices, and both (R2-sym) and the "no three parallel rungs" fact turn out to be necessary for it. We do not know whether $54/37$ is the limit of the method; the bottleneck is the near-resonant regime $\tau \geq \cos(\beta/2)$, where rhombi occur.

8. A layer bound and a new class of sets

In this section X is an arbitrary finite planar set with at least two distinct distances, $\Delta > \Delta_2$ are its two largest distances, and we scale so that $\Delta_2 = 1$. Every distance of X is at most 1 or equal to Δ. L_1 is the set of extreme points of X (vertices of $\operatorname{conv} X$), L_2 is the set of extreme points of $X \setminus L_1$, and $m_3 := n - |L_1| - |L_2|$. Let G be the graph on $L_1 \cup L_2$ whose edges are the Δ_2-pairs, let G' be its 2-core (repeatedly delete vertices of degree less than 2), and write $L_i' := L_i \cap V(G')$ and $\deg' = \deg_{G'}$. For $p \in X$ let $h(p)$ be the number of points of X at distance Δ_2 from p.

Lemma 8.1 (location)**.** *(i) Both endpoints of every Δ-pair lie in L_1.*
(ii) Every Δ_2-pair has an endpoint in L_1 and both endpoints in $L_1 \cup L_2$.
In particular G contains every Δ_2-pair of X.

Proof. (i) If $|pq| = \Delta$ then $X \subseteq \mathrm{B}(q, \Delta)$ and p lies on its boundary circle, so p is an extreme point of X. (ii) Let $|pq| = 1$. If $p \notin L_1$, then p has no Δ-partner by (i), so $X \subseteq \mathrm{B}(p, 1)$ and q, lying on the boundary circle, is in L_1. So some endpoint, say p, lies in L_1. Suppose $q \notin L_1$. Every point x with $|px| > 1$ has $|px| = \Delta$ and so lies in L_1. Hence $X \setminus L_1 \subseteq \mathrm{B}(p, 1)$ with q on the boundary circle, so q is extreme in $X \setminus L_1$, i.e. $q \in L_2$. □

Proposition 8.2 (core degree of L_2)**.** *Every $q \in L_2'$ has $\deg'(q) = 2$.*

Proof. Put $q = 0$. Since q is not extreme it has no Δ-partner (Lemma 8.1(i)), so $X \subseteq \mathrm{B}(0, 1)$ and the G-neighbours of q lie on the unit circle. Since $q \in V(G')$, $\deg'(q) \geq 2$. Suppose v_1, v_2, v_3 are three distinct G'-neighbours of q. Each $|v_i v_j|$ is either at most 1 or equal to Δ. The argument follows the three cases of Vesztergombi's inner-point analysis [19, pp. 192–194], with one change: we delete no points, and the contradiction in Cases 1 and 2 is with membership in the 2-core, via $\deg_G \geq \deg'$.

Case 3: at most one pair $v_i v_j$ at distance Δ. Label so that $|v_1 v_2|, |v_2 v_3| \leq 1$. Then v_2 lies on the shorter arc between v_1 and v_3, and in polar angle we may take v_2 at 0, v_1 at a and v_3 at $-b$ with $0 < a, b \leq 60°$. Since q is not extreme, the closed half-plane $\{x : x \cdot v_2 \leq 0\}$ contains a point $t \in X$, $t \neq q$. For $t \neq 0$ with $t \cdot v_2 \leq 0$ we get $|t - v_2|^2 = |t|^2 - 2t \cdot v_2 + 1 > 1$, so $|tv_2| = \Delta$. By symmetry assume the polar angle φ of t lies in $[90°, 180°]$. Then the angle between t and v_3 is $\varphi + b \in (90°, 240°]$, so $t \cdot v_3 < 0$ and $|tv_3| = \Delta$. Hence t lies on the perpendicular

bisector of v_2v_3, which passes through 0, so $\varphi = 180^\circ - b/2$. The angle between t and v_1 is then $180^\circ - b/2 - a \geq 90^\circ$, so $|tv_1|^2 \geq |t|^2 + 1 > 1$ and $|tv_1| = \Delta$. Now v_1, v_2, v_3 lie on both $\mathrm{C}(0,1)$ and $\mathrm{C}(t,\Delta)$, two distinct circles, which meet in at most two points. Contradiction. (This case uses only that q is not extreme.)

Case 2: exactly two pairs at Δ. Say $|v_1v_2| = |v_1v_3| = \Delta$ and $|v_2v_3| \leq 1$. Then v_1 lies on the perpendicular bisector of v_2v_3, and we may take $v_1 = (-1, 0)$, $v_{2,3} = (\cos a, \pm \sin a)$ with $0 < a \leq 30^\circ$ and $\Delta = 2\cos(a/2)$. We claim v_1 has no Δ_2-neighbour $r \neq q$; then $\deg_G(v_1) = 1$, contradicting $v_1 \in V(G')$. Such an $r = (x, y)$ satisfies $|r - v_1| = 1$ (so $|r|^2 = -2x$), $|r| \leq 1$, and $|r - v_i| \in [0, 1] \cup \{\Delta\}$ for $i = 2, 3$; by symmetry $y \geq 0$. Write $c = \cos a$, $s_a = \sin a$.

- *Both* $|r - v_2|, |r - v_3| \leq 1$. Impossible: $|r - v_3|^2 - 1 = -2x(1 + c) + 2ys_a \geq 0$, with equality only if $r = q$.
- $|r - v_2| = \Delta$ (the case $|r - v_3| = \Delta$ is symmetric). Using $|r|^2 = -2x$, this condition is the linear equation $-2x(1 + c) - 2s_ay = 1 + 2c$. Intersecting with the circle $|r - v_1| = 1$ gives $x = \big(\pm s_a\sqrt{8c + 7} - 3(1 + c)\big)/(4(1 + c))$, so $|r|^2 = \big(3(1 + c) \mp s_a\sqrt{8c + 7}\big)/(2(1 + c))$. The root with $+s_a\sqrt{8c + 7}$ in $|r|^2$ has $|r|^2 \geq 3/2$, outside the disc. For the other root, $|r|^2 \leq 1 \iff 1 + c \leq s_a\sqrt{8c + 7} \iff 1 + c \leq (1 - c)(8c + 7) \iff c^2 \leq \frac{3}{4} \iff a \geq 30^\circ$. So only $a = 30^\circ$ remains, and there $|r - v_3|^2 = 3 + c - s_a\sqrt{8c + 7} = 2$, which lies strictly between 1 and $\Delta^2 = 2 + \sqrt{3}$. Forbidden.
- *Both equal* Δ. Then r lies on the x-axis, so $r \in \{(0, 0), (-2, 0)\}$, and $(-2, 0)$ is outside the disc.

Case 1: all three pairs at Δ. Then v_1, v_2, v_3 form an equilateral triangle with circumcentre q, and $\Delta = \sqrt{3}$. Let $P \neq q$ be a further Δ_2-neighbour of some v_i, so $|P - v_i| = 1$ and $|P| \leq 1$. It cannot have both other distances at most 1: $|P - v_i| = 1$ gives $|P|^2 = 2P \cdot v_i$, the two conditions give $P \cdot (v_i - v_j) \leq 0$ for $j \neq i$, and adding and using $\sum_j v_j = 0$ gives $3P \cdot v_i \leq 0$, so $P = q$. Hence $|P - v_j| = \sqrt{3}$ for some $j \neq i$. Solving exactly, there are six candidate points, two for each v_i, each with $|P|^2 = (9 - \sqrt{33})/6$ and at squared distance $(7 - \sqrt{33})/2 \approx 0.63$ from the third vertex, so each is a Δ_2-neighbour of exactly one v_i. Their pairwise squared distances are $(29 - 5\sqrt{33})/6 \approx 0.046$ for three disjoint pairs (each joining candidates of different vertices), and otherwise $4/3$, $(9 - \sqrt{33})/2 \approx 1.628$ or $(17 - \sqrt{33})/6 \approx 1.876$, all strictly inside the forbidden interval $(1, 3)$. So a set of candidates that can coexist in X has at most two elements, at most two of the v_i have a second Δ_2-neighbour, and some v_i has $\deg_G(v_i) = 1$, contradicting $v_i \in V(G')$.

With four or more neighbours apply the argument to any three. Hence $\deg'(q) \leq 2$, so $\deg'(q) = 2$. □

Lemma 8.3 (opposite-side lemma)**.** *Let $|pa| = |pb| = |br| = 1$ with $r \neq p$. Suppose a lies strictly on one side of the line pb and r does not lie strictly on that side. Then $|ar| > 1$, so $|ar| = \Delta$.*

Proof. Put $p = 0$, $b = (1, 0)$, $a = (\cos A, -\sin A)$ with $0 < A < \pi$, and $r = b + (\cos F, \sin F)$ with $0 \leq F \leq \pi$; $F \neq \pi$ as $r \neq p$. Then $|r - a|^2 - 1 = 2(1 + \cos F)(1 - \cos A) + 2\sin F \sin A$, where the first term is positive and the second nonnegative. □

Lemma 8.4. *Let p be an extreme point of X with Δ_2-neighbours $w_1, \ldots, w_k$ in angular order. Then $\deg_G(w_i) = 1$ for every $3 \leq i \leq k - 2$.*

Proof. Since p is extreme, some line through p has $X \setminus \{p\}$ strictly on one side. So the directions of the w_j from p are distinct, lie in an open half-plane, and any two differ by less than 180°; for $j < i < l$, w_j and w_l lie strictly on opposite sides of the line pw_i. Suppose $r \neq p$ and $|rw_i| = 1$. Put $p = 0$, $w_i = (1, 0)$. Then w_{i-2}, w_{i-1} lie strictly below the x-axis and w_{i+1}, w_{i+2} strictly above; by symmetry assume r is not strictly below. Write $w_{i-2} = e(-\alpha_1)$, $w_{i-1} = e(-\alpha_2)$ with $0 < \alpha_2 < \alpha_1 < 180^\circ$. By Lemma 8.3, $|rw_{i-2}| = |rw_{i-1}| = \Delta$, so r lies on the perpendicular

bisector of $w_{i-2}w_{i-1}$, the line through 0 in direction $-\gamma$ with $\gamma = (\alpha_1+\alpha_2)/2 \in (0°, 180°)$. As $r \neq 0$ is not below the axis, $r = \lambda e(180° - \gamma)$ with $\lambda > 0$. From $|r - w_i| = 1$ we get $\lambda(\lambda + 2\cos\gamma) = 0$, so $\lambda = -2\cos\gamma$, whence $\gamma > 90°$ and $\alpha_1 > 90°$. Then $|w_{i-2}w_i| = 2\sin(\alpha_1/2) > \sqrt{2} > 1$, so $|w_{i-2}w_i| = \Delta$. Let $\theta > 0$ be the direction of w_{i+1}; then $\alpha_1 + \theta < 180°$, and since the chord length $2\sin(x/2)$ is strictly increasing on $[0°, 180°]$, $|w_{i-2}w_{i+1}| = 2\sin((\alpha_1+\theta)/2) > 2\sin(\alpha_1/2) = \Delta$, contradicting the maximality of Δ. □

Theorem 8.5 (degree at most 4)**.** *Every $p \in L_1'$ has $\deg'(p) \leq 4$.*

Proof. Suppose $p \in L_1'$ has five G'-neighbours $u_1, \ldots, u_5$ in angular order. Among the G-neighbours of p, u_3 has at least two on each side, so $u_3 = w_i$ with $3 \leq i \leq k-2$, and $\deg_G(u_3) = 1$ by Lemma 8.4. But $u_3 \in V(G')$ forces $\deg_G(u_3) \geq 2$. □

Theorem 8.6 (layer bound)**.** *For every finite $X \subset \mathbb{R}^2$ with at least two distinct distances,*

$$\mu(\Delta_2) \;\leq\; |L_1| + |L_2| + \tfrac{1}{2}\sum_{p \in L_1'} (\deg'(p) - 2).$$

Proof. By Lemma 8.1, $\mu(\Delta_2) = e(G)$. Each deletion made while pruning G to its 2-core removes at most one edge, so $e(G) \leq e(G') + |V(G) \setminus V(G')|$. By Proposition 8.2, $e(G') = \frac{1}{2}\sum_{v \in V(G')} \deg'(v) = |V(G')| + \frac{1}{2}\sum_{p \in L_1'}(\deg'(p) - 2)$. Add the two estimates and use $|V(G)| = |L_1| + |L_2|$. □

Corollary 8.7. *Assume* (H3)*: no point of L_1 has four or more points of X at distance Δ_2. (This holds, for instance, if no four points of X are concyclic.) Then*

$$\mu(\Delta_2) \leq |L_1| + |L_2| + \tfrac{1}{2}\#\{p \in L_1' : \deg'(p) = 3\} \leq \tfrac{3}{2}|L_1| + |L_2|.$$

Consequently, under (H3)*, if $|L_1| \leq 2m_3$ then Δ and Δ_2 are two distinct distances each occurring at least once and at most n times.*

Proof. Under (H3), $2 \leq \deg'(p) \leq h(p) \leq 3$ for $p \in L_1'$. The condition $\frac{3}{2}|L_1| + |L_2| \leq n$ is equivalent to $|L_1| \leq 2m_3$, and $\mu(\Delta) \leq n$ by [14]. □

Corollary 8.8 (unconditional form)**.** *For every finite X with at least two distances, $\mu(\Delta_2) \leq |L_1| + |L_2| + \frac{1}{2}\sum_{p \in L_1}(\min\{h(p), 4\} - 2)^+$. Hence $\mu(\Delta_2) \leq n$ whenever $\sum_{p \in L_1}(\min\{h(p), 4\} - 2)^+ \leq 2m_3$.*

Proof. Combine Theorem 8.6 with $2 \leq \deg'(p) \leq \min\{h(p), 4\}$ on L_1' (Theorem 8.5). □

Remark 8.9. (a) Corollary 8.7 covers sets not covered by (1). For example, if $|L_1| = |L_2| = 0.4n$ and $m_3 = 0.2n$, the three quantities in (1) are $1.2n$, $1.33n$ and $1.2n$, while Corollary 8.7 gives $\mu(\Delta_2) \leq n$ (under (H3)). Of course (1) needs no hypothesis like (H3).

(b) Using $\deg' \leq 4$, Theorem 8.6 recovers the bound $2|L_1| + |L_2|$ of [5, Theorem 1.3]. Proposition 8.2 and Theorem 8.5 are the inputs from [20] (Propositions 3 and 5–8 there, as used in [5]) needed for this; we prove them directly in the 2-core, without deleting points. We have not checked the other inputs used in [5] for the first two bounds in (1).

(c) In Corollary 8.7, the bound $\frac{3}{2}|L_1| + |L_2|$ is attained under (H3) alone by the following example, which we have checked numerically for $m = 8, 12, 16$ (floating point): a regular m-gon with m even, with a point at distance Δ_2 from both endpoints of every other edge (CDL's "servers"). These examples contain concyclic quadruples, so they do not bear on Conjecture 10.1 below.

9. Sets in convex position

A set X is in *convex position* if every point of X is a vertex of $\operatorname{conv} X$; in particular no three points of X are collinear. Throughout this section $|X| = n$, $m := \lfloor n/2 \rfloor$, $N := \binom{n}{2}$, $D = D(X)$ is the number of distinct distances, and $k = k(X)$ is the number of rare distances ($1 \le \mu(d) \le n$). A distance is *frequent* if $\mu(d) \ge n+1$. R_N denotes the vertex set of a regular N-gon. By Hopf–Pannwitz, Δ is always rare; Clemen, Dumitrescu and Liu proved $k \ge 2$ for $n \ge 5$ [5, Theorem 1.2]. For $n = 5$ this is best possible: R_5 has $k = 2$.

Theorem 9.1. *Let X be a set of n points in convex position with $6 \le n \le 13$. Then at least three distinct distances each occur at least once and at most n times. The same holds for every even $n \ge 6$.*

The cases have different status:

- $n = 6, 7, 8$: a corollary of published results (Proposition 9.3).
- even $n \ge 10$: Theorem 9.4, a parity-sharp version of the proof of [5, Theorem 1.2];
- $n = 9$: Theorem 9.5, from the classification of Erdős and Fishburn [10];
- $n = 11, 13$: Theorem 9.8, new and computer-assisted. The $n = 7$ case is re-proved the same way, without the literature.

9.1. Counting and the literature.

Lemma 9.2 (parity-sharp counting)**.** *$(D-k)(n+1)+k \le N$. Consequently, if $n = 2m+1$ and $D = m+K$ with $1 \le K \le m+1$, then $k \ge K+1$; if $n = 2m$ and $D = m+K$ with $1 \le K < 2m$, then $k \ge K+2$.*

Proof. The first inequality holds because frequent distances contribute at least $n+1$ pairs each and rare ones at least 1. It gives $k \ge (D(n+1) - N)/n$. For $n = 2m+1$ the right side is $K + (m+K)/(2m+1)$, with $0 < (m+K)/(2m+1) \le 1$. For $n = 2m$ it is $K + 1 + K/(2m)$, with $0 < K/(2m) < 1$. □

We use the following results from the literature.

(Alt) Altman [1, 2]: $D \ge m$; for odd n, $D = m$ only for R_n (see also [13, Theorem 1]).

(Fi) Fishburn [13]: for even $n \ge 8$, $D = n/2$ if and only if X is R_n or n vertices of R_{n+1}. For $n = 6$ the hexagons with $D = 3$ are R_6, R_7 minus a vertex, and one further hexagon [13, Theorem 2], [8], with multiplicity profiles $(6,6,3)$, $(5,5,5)$ and $(6,6,3)$ respectively (as recorded in [5, §2]; we re-checked the profiles in exact arithmetic).

(EF96) Erdős–Fishburn [10]: a convex nonagon has exactly 5 distances if and only if its vertices are 9 vertices of R_{10} or of R_{11}.

(r_2) Erdős–Fishburn [8, §3], quoting Fishburn's technical report [12]: the largest possible second-largest multiplicity in a convex n-gon is $r_2^{(n)} = 1, 3, 5, 6, 7, 8$ for $n = 3, \dots, 8$.

A set on a circle has every distance class of maximum degree at most 2, hence of size at most n; so for cocircular X all distances are rare and $k = D$.

Proposition 9.3 (from the literature)**.** *For $n = 6, 7, 8$ in convex position, $k \ge 3$.*

Proof. By (r_2), $r_2^{(n)} \le n$ for $n \le 8$, so at most one distance is frequent and $k \ge D - 1$. By (Alt), $D \ge 3$ for $n = 6, 7$ and $D \ge 4$ for $n = 8$. So $k \ge 3$ unless $n \in \{6, 7\}$ and $D = 3$. For $n = 7$, $D = 3$ forces R_7 (profile $(7,7,7)$, $k = 3$); for $n = 6$, (Fi) gives the profiles $(6,6,3)$ and $(5,5,5)$, so $k = 3$. □

The value $r_2^{(7)} = 7$, and the other values of $r_2^{(n)}$, rest on an unrefereed technical report [12] quoted in a refereed paper [8]; for $n = 7$ we give an independent proof in Theorem 9.8.

Theorem 9.4 (even n)**.** *Let X be in convex position with $n = 2m \geq 6$. Then $k \geq 3$. If $n \geq 8$ and $k \leq 3$, then $D = m + 1$.*

Proof. By (Alt), $D \geq m$. If $D \geq m+1$, Lemma 9.2 gives $k \geq 3$, and $k \geq 4$ if $D \geq m+2$. If $D = m$ and $n = 6$, (Fi) gives $k = 3$. If $D = m$ and $n \geq 8$, then X is R_n, with profile $(n, \ldots, n, n/2)$, or n vertices of R_{n+1}, with profile $(n-1, \ldots, n-1)$; in both cases $k = m \geq 4$. □

Theorem 9.4 is essentially the argument of [5, Theorem 1.2], with the counting done with parity. It adds to the literature only for even $n \geq 10$.

Theorem 9.5 ($n = 9$)**.** *Every set of 9 points in convex position has $k \geq 3$.*

Proof. Here $m = 4$. If $D = 4$, then $X = R_9$ by (Alt) and $k = 4$. If $D = 5$, then X is cocircular by (EF96) and $k = D = 5$. If $D \geq 6$, Lemma 9.2 gives $k \geq 3$. □

9.2. $n = 7, 11, 13$**: an ordinal reduction and SAT certificates.** Label $X = (p_0, \ldots, p_{n-1})$ in convex cyclic order, let $d_1 < \cdots < d_D$ be the distinct distances, and colour each pair $\{i, j\}$ with $c(ij) = t$ where $|p_i p_j| = d_t$.

Lemma 9.6 (quadrilateral inequality)**.** *For $i < j < k < l$ (in cyclic order), $|p_i p_k| + |p_j p_l| > |p_i p_j| + |p_k p_l|$ and $|p_i p_k| + |p_j p_l| > |p_j p_k| + |p_l p_i|$.*

Proof. The diagonals of the convex quadrilateral $p_i p_j p_k p_l$ meet at a point O that lies on neither of the sides $p_i p_j$, $p_k p_l$. So $|p_i p_k| + |p_j p_l| = (|p_i O| + |O p_j|) + (|p_k O| + |O p_l|) > |p_i p_j| + |p_k p_l|$, by the strict triangle inequality. The other inequality is the same. □

Ordinal form. Let $x_1 \leq x_2$ be the two diagonal colours, and $y_1 \leq y_2$ the colours of one pair of opposite sides. Then it is *not* the case that $x_1 \leq y_1$ and $x_2 \leq y_2$, since otherwise $|p_i p_k| + |p_j p_l| \leq$ the sum of the two sides. We call a colouring of the pairs of an n-cycle's vertex set satisfying this for all 4-tuples and both pairs of opposite sides a *Q-colouring*. This ordinal use of the quadrilateral inequality goes back to Altman [1]. Abstracting a convex distance problem to a finite search in this way is the method of Morić and Pritchard [17], whose Fact 3.1 is Lemma 9.6.

Some of our certificates also use a second valid constraint (*Lemma T*): if $w \neq w'$ lie on the same open boundary arc between p_i and p_j, then $(c(iw), c(jw)) \neq (c(iw'), c(jw'))$, because a point is determined by its distances to p_i, p_j and a side of the line $p_i p_j$.

Lemma 9.7 (reduction to profiles)**.** *If $k \leq 2$, then at least $D - 2$ colour classes have size at least $n + 1$, and $D \leq D_{\max}(n)$, the largest D with $(D-2)(n+1) + 2 \leq N$. For $n = 7, 11, 13, 15$, $D_{\max} = 4, 6, 7, 8$.*

Call a Q-colouring using exactly D colours, all of them, with at least F classes of size at least $n + 1$, an (n, D, F) *ordinal counterexample*. Since Lemma 9.6 (and Lemma T) are necessary for geometric realisability, the following suffices to prove $k \geq 3$ for a given n: *for every $2 \leq D \leq D_{\max}(n)$ there is no $(n, D, D-2)$ ordinal counterexample.* (We check the smaller D even where (Alt) would exclude them, so that no citation is needed; for $D \leq 4$ we check the stronger statement with $F = 0$, i.e. no Q-colouring with D colours at all.) No realisation step and no floating-point arithmetic enters.

Theorem 9.8. *Every set of n points in convex position with $n \in \{7, 11, 13\}$ has at least three rare distances. For these n, the ordinal quadrilateral inequality (Lemma 9.6) alone implies the statement.*

Proof (computer-assisted). By Lemma 9.7 it suffices that the following instances have no ordinal counterexample:

- $n = 7$: $(7, 2, 0)$, $(7, 3, 1)$, $(7, 4, 2)$;

- $n = 11$: $(11, 2, 0)$, $(11, 3, 0)$, $(11, 4, 0)$, $(11, 5, 3)$, $(11, 6, 4)$;
- $n = 13$: $(13, 2, 0)$, $(13, 3, 0)$, $(13, 4, 0)$, $(13, 5, 0)$, $(13, 6, 4)$, $(13, 7, 5)$.

Each is encoded in CNF and shown unsatisfiable, with the refutation checked as described below and in Appendix A. □

Encoding. Variables $x_{e,c}$ say that pair e has colour c; there are exactly-one constraints per pair and at-least-one constraints per colour. In the *direct* encoding of Lemma 9.6, each dominated 4-tuple of colours is forbidden. In the *order* encoding, auxiliary literals $y_{e,t} \Leftrightarrow c(e) \geq t$ are introduced and "$c(a) \leq t \leq c(b)$ and $c(a') \leq u \leq c(b')$" is forbidden for both diagonal/side matchings; this is equivalent, since sorted domination holds if and only if some matching dominates. Selector variables f_c imply "class c has at least $n+1$ pairs" through sequential counters, and $\sum_c f_c \geq F$.

Certification.

- *DRAT/LRAT.* For every instance with $n = 11$ and $n = 13$ listed above, and for $(7, 4, 2)$, a DRUP proof produced by Glucose 4 [3] was verified by `drat-trim` [21], which also emitted an LRAT proof that was re-verified by `lrat-check` [6]. The largest certificate, for $(13, 7, 5)$, has 446,686 lemmas (a 37 MB DRUP proof and an 82 MB LRAT proof). For $n = 13$ all instances were certified in the Q-only encoding. For $n = 11$, $(11, 6, 4)$ was certified in the Q-only encoding (35,510 lemmas), and $(11, 2..5, \cdot)$ in the encoding with Q and Lemma T (Q-only is covered by the Lean proof below). A truncated proof is correctly rejected. On this route, the trusted components are the CNF encoder (a short Python program, cross-validated as in Appendix A) and the proof checkers.
- $n = 7$ *without SAT.* A plain backtracking enumeration of all colourings, with Q checked as soon as its pairs are assigned, finds no ordinal counterexample for $D = 2, 3, 4$. For $D = 4$ it does so with Lemma T switched off.
- *Lean 4.* The statements for $n = 7$, 11 and 13 are formalised in Lean 4 [7] with Mathlib [18]; see below.

The Lean formalisation ($n = 7$, 11, 13). The file `Erdos/Erdos/E132Convex.lean` proves, without `sorry`,

```
theorem erdos132_convex7 (p :  Fin 7 → EuclideanSpace ℝ (Fin 2)) (hp :
          StrictConvexPolygon p) :  3 ≤ (rareDists p).card
```

and the analogous `erdos132_convex11` and `erdos132_convex13` (file `E132Convex13.lean`) for `Fin 11` and `Fin 13`. Here `StrictConvexPolygon p` means that *every* index-increasing triple (p_i, p_j, p_k), $i < j < k$, is positively oriented; that is, the vertices are given as a strictly convex polygon listed counter-clockwise. `rareDists p` is the set of distances realised by pairs $i < j$ with multiplicity at most n. The proof formalises Lemma 9.6 (via the intersection point of the diagonals and strict convexity of the Euclidean norm), the passage to ranks and Q-colourings, Lemma 9.7, and the correctness of a bit-vector encoding of the finite statement "every Q-colouring with colours below $D_{\max}$ has at least three classes of size in $[1, n]$". This covers all $D \leq D_{\max}$ at once, with unused colours allowed. The finite statement is closed by Lean's `bv_decide`, which bit-blasts the formula with a verified bit-blaster, calls the bundled SAT solver CaDiCaL, and checks the resulting LRAT certificate with a checker that is proved correct in Lean.

Trust base. `#print axioms` reports, for each theorem, the standard axioms `propext`, `Classical.choice`, `Quot.sound`, plus one axiom generated by `bv_decide` (`..._native.bv_decide.ax_1_14`). That axiom asserts that the verified LRAT checker `verifyBVExpr`

returns `true` on the concrete formula and certificate. The value `true` is obtained by running compiled code, not by kernel reduction, so the trust base is comparable to that of `native_decide`: it includes the Lean compiler and runtime. Neither our Python encoder, nor any external SAT solver, nor `drat-trim` is trusted on this route. An attempt to check the $n = 7$ axiom statement by kernel reduction ran for more than 20 minutes without finishing and was stopped.

Unordered point sets. The file `E132ConvexFinset.lean` proves (using only the three standard axioms) that a finite set S in convex position, in the sense of Mathlib's `ConvexIndependent`, can be listed as a `StrictConvexPolygon`. The proof sorts $S \setminus \{p_0\}$ by orientation about a point $p_0 \in S$; a negatively oriented increasing triple would put a point of S inside a triangle spanned by others. It also proves that the rare distances of S (unordered pairs) coincide with `rareDists p` for any such listing. This gives `erdos132_convex{7,11,13}_finset`: for `S : Finset` with `S.card` $= n$ and `ConvexIndependent`, $3 \le$ `(rareDistsS S).card`.

The layer bound. The results of Section 8 are formalised in `E132Layer.lean` without `sorry`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` (no `bv_decide` or `native_decide`). This covers the localisation of Δ_2-pairs to $L_1 \cup L_2$, the degree-2 property of L_2-vertices in the 2-core, the bound deg ≤ 4 on hull vertices, Theorem A, Corollary A1 and its unconditional form. There L_1 is the set of extreme points of X, shown equal to the vertex set of conv X, and the 2-core is the largest subset with minimum degree at least 2.

The main theorem. Theorem 1 is formalised in full in `E132Main*.lean`, `E132ConvexCurve.lean` and `E132Ve87.lean` (about 7,000 lines):

```
theorem erdos132_main : ∃ C₀ : ℝ, ∀ X : Finset Pt, 2 ≤ X.card → (min (mult
    X (dist2 X)) (mult X (minDist X)) : ℝ) ≤ 54/37 * X.card + C₀
```

with `mult X d` the number of unordered pairs of distinct points of X at distance d, and `dist2`, `minDist` the second-largest and smallest distance. There is no `sorry`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The formalisation also proves Vesztergombi's bound $\mu(\Delta_2) \le \frac{3}{2}n$ and the Hopf–Pannwitz bound, so no result is used as a black box. In three places it takes a different route from the text: (a) Vesztergombi's bound is proved in the 2-core (every degree-4 core vertex has two degree-2 core neighbours, then double counting), with no point deletion; (b) the bad-edge bounds of Section 4 are obtained by counting over directions of outward normals instead of arc length, so Lemma 5.2 (lens arc length) is not needed; (c) the degenerate-regime incidence bound uses a local charging argument, and the mixed-T bound of Lemma 7.1 is proved in a weaker form (at most 1200 instead of 71), which only changes the constant C_0.

Limitations of the formalisation. (i) The `bv_decide` axiom described above remains. (ii) Only the $n = 7$ LRAT certificate (1.6 MB) is stored and replayed with `bv_check`; for $n = 11, 13$ the certificates (23 MB and 97 MB) are not stored, so compilation re-runs CaDiCaL (about 40 s and 4 min respectively). (iii) The paper's Lemma 5.2 is not formalised; the formal proof of the main theorem does not need it (see above).

Remark 9.9. (a) The method is tight at "three": $(11, 6, 3)$ is satisfiable (e.g. with class sizes $(12, 9, 12, 12, 6, 4)$), as is $(9, 5, 2)$.

(b) The ordinal statement fails for $n = 5, 6$: $(6, 3, 1)$ has an ordinal solution with profile $(9, 3, 3)$, so at $n = 6$ geometry beyond Lemma 9.6 is needed.

(c) For $n = 15$, all instances with $D \le 7$ were found unsatisfiable by CaDiCaL (solver output only, not certified), but the hard instance $(15, 8, 6)$ was unresolved after more than 20 minutes. So $n = 15$ is open.

(d) Theorem 9.8 for $n = 7$ also follows from the literature (Proposition 9.3).

Remark 9.10 (related work). Independently, a Lean project on the Lean Pool [16] proves that for point sets with coordinates in certain restricted fields (e.g. $\mathbb{Q}(\sqrt{2})$), $n \le 2^r$ where r is the number of rare distances. This gives at least three rare distances for $n \ge 8$ in those coordinate fields; it neither implies nor is implied by Theorem 9.1. Other contributions on the forum [11] concern the first question of Problem #132 (at least two rare distances) for small n and general sets.

10. Open problems

Problem 1.6 of [5]**.** We now know $9/7 \le L \le 54/37$, i.e. $1.2857\ldots \le L \le 1.4594\ldots$. The lower bound construction [15] is conjectured there to be optimal. In our proof the constant $54/37$ comes only from the near-resonant regime $\tau \ge \cos(\beta/2)$ (the diameter exceeds Δ_2 by almost exactly δ); every other regime gives at most $36/25$. Improving the upper bound would therefore require a better treatment of that regime, for instance a bound on $\mu(\Delta_2)$ that uses the structure of S there rather than only $\mu(\Delta_2) \le \frac{3}{2}|S|$.

Genericity.

Conjecture 10.1. *If no four points of X are concyclic, then $\mu(\Delta_2) \le n + O(1)$. More strongly, the number of vertices of degree 3 in the 2-core G' is $O(1)$.*

Together with Theorem 8.6, the stronger form would answer the first question of Problem #132 for sets with no four concyclic points and with at least a constant number of points of depth at least 3. An elementary observation, probably folklore, points in this direction: if $|ab| = |cd| = \Delta$ and $|ac| = |bd| = \Delta_2$, then a, b, c, d are concyclic. (The triangles abc and dcb are congruent. A half-turn would make $abdc$ a parallelogram with $|ad|^2 + |bc|^2 = 2(\Delta^2 + \Delta_2^2) > 2\Delta^2$, which is impossible, so $abdc$ is an isosceles trapezoid.) The known extremal configurations for $\mu(\Delta_2)$ contain such alternating 4-cycles. We have only weak numerical evidence for Conjecture 10.1 (a greedy search).

Convex position.

Conjecture 10.2 (Q-ORD)**.** *For every $n \ge 7$, every Q-colouring of the pairs of $\{0, \ldots, n-1\}$ has at least three colour classes of size at most n.*

Q-ORD holds for $7 \le n \le 13$ (Theorem 9.8), and it would imply that every set of $n \ge 7$ points in convex position has at least three rare distances. For $n = 15$ only $(15, 8, 6)$ remains. The solving time grows roughly tenfold for each increase of n by 2, so a proof for all n must be human. Deletion-minimal refutations are not local ($(7, 4, 2)$ already needs 25 of the 35 quadrilaterals), so such a proof will have to be global.

For odd $n = 2m + 1 \ge 7$ (no computation needed), Lemma 9.2 and (Alt) show that a convex set with $k \le 2$ must have $D = m + 1$, $k = 2$ and $m - 1$ frequent distances. Moreover, writing the frequent multiplicities as $2m + 2 + e_d$ and the two rare ones as a, b, the identity $a + b + \sum_d e_d = m + 2$ holds. (Indeed $D = m$ gives R_n with $k = m \ge 3$, $D \ge m + 2$ gives $k \ge 3$, and the identity is $(m-1)(2m+2) + \sum e_d + a + b = m(2m+1)$.) So for odd $n \ge 15$ the question in convex position is about this single kind of configuration. Beyond three, the second question of Problem #132 asks whether $k \to \infty$; this is open even in convex position.

Appendix A. Computational verification

All scripts are in the directory `problems/132/scripts/` of the accompanying repository **[REPOSITORY URL]**. None of the computations below is used as a proof step, except the SAT/LRAT certificates and the Lean proof for Theorem 9.8. The rest are independent checks of hand-proved steps. Tags [Ck] in the text refer to sections of `writeup_a2_checks.py`.

Theorem 1.

- `writeup_a2_checks.py`: 58 assertions in exact rationals and rational interval arithmetic, with one-sided rational bounds for $\sqrt{\cdot}$, π, sin, cos, arcsin. Sections: C1 the LP certificates of Lemmas 3.3 and 3.4; C2 the identity of Lemma 2.2(c) and the relation between τ and t; C3 small-angle facts for $\Delta_2 \geq 10^4$; C4 the Region I windows; C5 the Region II constants ((R1), (R2), (R2-sym), mixed T-edges, rhombus threshold, pairing ratio); C6 the corner lemma; C7 the lens constants; C8 case (c2); C9 the bad-edge totals; C10 the degenerate-regime constants; C11 the small-case packing bound. All pass.
- `ref2_numeric.py`: an independent set of 40 exact assertions (second referee pass) covering the same inequalities, together with an adversarial floating-point search for (R1), (R2), (R2-sym). The search is reported, not asserted. Its maxima were 0.906, 0.897 and 0.914, against the proved bounds 0.908, 0.918 and 0.948.
- `ref2_ownership.py`: an exhaustive exact check of the ownership/pairing bookkeeping in Section 5, in the abstract model the argument uses: rows that are disjoint unions of paths and cycles of length ≥ 3, rungs forming a matching, (R2), (R2-sym), and no two consecutive rhombi. It covers all row structures up to isomorphism with at most 6 vertices per row and all partial rung matchings. There were 2,365,426 admissible configurations without rhombi, with maximum ratio exactly 5/4, and 2,741,183 with rhombi allowed, with maximum ratio exactly 4/3. Mutation tests show that (R2-sym) and the rule excluding two consecutive rhombi are both needed.
- `ref_degen.py`, `degen_check.py`, `ref2_degen.py`: adversarial searches in the degenerate regime, including 5500 configurations built so that three or more circles $\mathrm{C}(w, \Delta_2)$ pass through one vertex of ∂K. The maximum of $\mu(\Delta_2) - n$ found was 0, against the proved bound 1.
- `ref_checks.py`, `a2_checks.py`: random tests of the lens lemma (about 7000 lenses, worst ratio 0.66) and of the exact single-centre strip in the near-resonant regime.

The whole proof was also re-derived by hand twice, independently, in adversarial "referee" passes. Vesztergombi's theorem [19] was checked in the original: its main theorem is $\mu(\Delta_2) \leq \frac{3}{2}n$ for all n, with no additive constant.

Section 8.

- `va1b_exact.py` (sympy, mpmath interval arithmetic): exact computation of the six Case 1 candidates of Proposition 8.2 and their pairwise squared distances, with an exhaustive check that no compatible set of candidates covers all three v_i. It also gives a closed form and a rigorous interval check (6000 subintervals, 0 failures) of the Case 2 branch $|r - v_2| = \Delta$.
- `va1c_sympy.py`: symbolic checks of the identity in Lemma 8.3, of $\lambda(\lambda + 2\cos\gamma) = 0$, of the chord formula, and of the bisector direction in Lemma 8.4.
- `va1c_local.py`: exhaustive search for the configuration of Lemma 8.4 with unit vectors from $\mathbb{Z}[\omega]$ (norm ≤ 147, exact), $\mathbb{Z}[i]$ (norm ≤ 325, exact) and N-th roots of unity ($5 \leq N \leq 60$): 0 hits in about $3.2 \cdot 10^6$ configurations. A control run with weakened hypotheses does find hits.
- `va1_clique.py`, `va1_cyclo.py`, `va1c_random.py`: searches over maximal compatible sets in lattice and cyclotomic pools, plus greedy floating-point constructions. Every set was tested against Lemma 8.1, Proposition 8.2, Theorem 8.5, Theorem 8.6 and Corollaries 8.7 and 8.8: more than 800,000 sets in the first searches and about 81,000 in the second, with 0 violations.

Section 9.

- `ver_classification_k.py`, `ver_hexagons.py`: exact chord-index profiles for the classified extremal sets used in Theorems 9.4 and 9.5 and Proposition 9.3, and strict convexity and multiplicity profiles of the three 3-distance hexagons.
- `counting_parity.py`: the closed forms of Lemma 9.2 for $n < 400$.
- `b3_sat.py`: the CNF encoder (direct and order encodings, with or without Lemma T). `b3_enum.py`: the standalone backtracking enumerator for $n = 7$.
- `b3_validate.py`: encoder validation. Every 7-subset of R_N, $7 \le N \le 16$ (11,440 sets), and every 11-subset of R_N, $N \le 15$, satisfies all constraints under exact chord-index ranks, as do 3000 (for $n = 7$) and 500 (for $n = 11$) random convex polygons. The $(7, 3, 0)$ enumeration returns exactly the R_7 colouring. Every satisfying assignment printed by the SAT route was re-checked against the independent constraint code of `b3_enum.py`.
- Four solvers (CaDiCaL, Glucose 4, MiniSat 2.2, Lingeling) agree on the $n = 7$ and $n = 11$ instances.
- `b3_drup_check.py`: an independent forward RUP checker, used for the $n = 7$, $n = 9$ and $n = 11$ DRUP proofs. Its self-test rejects an empty proof of a satisfiable instance, a proof with one negated lemma, and a truncated proof.
- `b3_drat_trim.py`: regenerates each CNF and Glucose proof deterministically and runs `drat-trim` with LRAT output, followed by `lrat-check`. The CNF, DRUP and LRAT files are not stored in the repository because of their size.

instance (n, D, F)	encoding	lemmas	`drat-trim`	`lrat-check`
$(7, 4, 2)$	Q	714	verified	verified
$(11, 2, 0)$, $(11, 3, 0)$, $(11, 4, 0)$	Q+T	3, 8, 56	verified	verified
$(11, 5, 3)$	Q+T	524	verified	verified
$(11, 6, 4)$	Q	35,510	verified	verified
$(13, 2, 0)$, $(13, 3, 0)$, $(13, 4, 0)$, $(13, 5, 0)$	Q	2, 7, 82, 448	verified	verified
$(13, 6, 4)$	Q	8,463	verified	verified
$(13, 7, 5)$	Q	446,686	verified	verified

TABLE 2. Certified refutations for Theorem 9.8.

Lean. The whole development (all `E132*.lean` files: the convex-position results, the layer bound, Vesztergombi's and Hopf–Pannwitz's bounds, and the main theorem) builds with `lake exe cache get` followed by `lake build`; on a laptop this takes about 6 minutes after the Mathlib cache is downloaded, of which about 4 minutes is `bv_decide` for $n = 13$. The toolchain is Lean `v4.35.0-rc2` with Mathlib.

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